Fluid-Dynamic Stability Filtering for Zero-Noise Extrapolation in NISQ-Era Quantum Circuits

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Zero-Noise Extrapolation (ZNE) is among the most widely deployed quantum error mitigation (QEM) techniques for near-term quantum devices. However, the choice of noise scaling schedule remains largely empirical, and no standard method exists for validating whether a given ZNE extrapolation is physically reliable or has diverged into an artifact. We introduce two contributions to address this gap. First, a *fluid-dynamic stability filter* derived from the Navier-Stokes energy dissipation norm, $\varepsilon = \nu \int_\Omega \|\nabla u\|^2 \, d\Omega < \Gamma$, applied to ZNE extrapolation residuals as a post-hoc reliability criterion. Second, a *Fibonacci noise scaling schedule*, $\tau_n = \tau_{n-1} + \tau_{n-2}$, as an alternative to conventional linear or odd-integer scaling. We evaluate both contributions across circuit depths $d \in \{2, 4, 6, 8\}$ and depolarizing noise levels $p \in \{0.001, 0.005, 0.01, 0.02, 0.05\}$ using shot-based density matrix simulation in Cirq. Our results across 200 configurations (4 depths $\times$ 5 noise levels $\times$ 10 trials) show that all three schedules remain well below the stability threshold $\Gamma = 0.05$ under standard conditions, validating the filter as a reliable diagnostic. The Fibonacci schedule exhibits a higher mean energy norm ($\varepsilon = 0.00219$) than linear ($\varepsilon = 0.00175$) or odd ($\varepsilon = 0.00095$) at depth $d = 4$, $p = 0.01$, revealing a coverage-smoothness tradeoff introduced by its wider scale factor spacing. Code is available at GitHub and archived at Zenodo.

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INTRODUCTION

Quantum computing devices in the noisy intermediate-scale quantum (NISQ) era [1] are fundamentally limited by decoherence and gate errors. Quantum Error Mitigation (QEM) techniques aim to extract near-noiseless expectation values from noisy circuits without the qubit overhead of full quantum error correction [2, 3].

Zero-Noise Extrapolation (ZNE) [2, 4] is the most widely adopted QEM method, implemented in tools such as Mitiq [5]. ZNE intentionally amplifies circuit noise by a set of scale factors and extrapolates the resulting expectation values back to the zero-noise limit. Despite its widespread use, two fundamental questions remain open:

1. **Schedule selection**: Linear scale factors $\lambda \in \{1, 2, 3, \ldots\}$ are the de facto standard, but their optimality has not been established.

2. **Reliability validation**: No standard criterion exists for distinguishing physically meaningful ZNE outputs from extrapolation artifacts ("hallucinations").

We address both questions. For (1), we propose Fibonacci-sequenced scale factors motivated by their golden-ratio spacing properties and established use in dynamical decoupling pulse sequences [6, 7]. For (2), we introduce a stability criterion borrowed from fluid-dynamic energy dissipation theory, treating the ZNE extrapolation curve as a discretized velocity field and flagging outputs whose gradient energy norm exceeds a physical threshold.

RELATED WORK

Zero-Noise Extrapolation Variants

Since its introduction by Temme et al. [2] and Li and Benjamin [4], ZNE has been extended in several directions. Giurgica-Tiron et al. [8] introduced digital noise scaling via unitary folding and studied multiple extrapolation methods including Richardson extrapolation, demonstrating error reductions of up to 24$\times$ over non-mitigated circuits. Krebsbach et al. [9] analyzed the optimization of Richardson extrapolation specifically for QEM, establishing theoretical bounds on its performance. Most recently, direct polynomial analysis of ZNE with explicit error bounds was provided by [10], offering the most rigorous theoretical treatment to date.

Crucially, none of these works address the *reliability* of a given ZNE output — the question of whether a specific extrapolation has converged to a physically meaningful value or diverged into an artifact. Our stability filter addresses this gap directly.

Noise Scaling Schedule Selection

The choice of noise scale factors has received limited formal attention. Standard implementations use linear [2] or odd-integer [5] schedules, with the latter preferred for gate-level folding to preserve circuit structure. A noise-aware folding method was proposed by [11], which redistributes noise amplification based on hardware characteristics, representing the closest prior work to our schedule investigation. However, no prior work has

systematically compared schedule spacing distributions or proposed golden-ratio-motivated schedules for ZNE.

ZNE on Hardware and Logical Qubits

ZNE has been demonstrated experimentally on superconducting qubits [12], silicon spin qubits [13], and quantum annealing platforms [14], confirming its broad applicability across qubit modalities. Our work focuses on simulation-level validation of scheduling and filtering methods, complementary to these hardware demonstrations.

Physics-Informed Methods in Quantum Computing

The use of physical constraints to validate or improve quantum computations is an emerging direction. Physics-informed neural networks [15] have been applied to quantum simulation problems, and stability criteria from fluid dynamics have been used as convergence monitors in classical numerical methods. Our work is the first, to our knowledge, to apply Navier-Stokes energy dissipation norms as a post-hoc reliability criterion for quantum error mitigation outputs.

BACKGROUND

Zero-Noise Extrapolation

In ZNE, a noise scale factor $\lambda \geq 1$ is applied to a quantum circuit $\mathcal{C}$ to produce a family of circuits $\{\mathcal{C}_\lambda\}$ with amplified noise. Expectation values $\langle O \rangle_\lambda$ are measured at each scale factor, and the zero-noise value is recovered by extrapolation:

$$\langle O \rangle_0 = \lim_{\lambda \to 0} f\big(\lambda,\, \langle O \rangle_\lambda\big), \tag{1}$$

where f is typically a polynomial or Richardson extrapolant.

The standard linear schedule uses $\lambda \in \{1, 2, 3, \ldots, K\}$. Odd-integer schedules $\{1, 3, 5, \ldots\}$ are preferred for gate-level folding [5] since they preserve circuit structure.

Fibonacci Sequences in Quantum Control

Fibonacci-based pulse sequences have been studied in the context of dynamical decoupling, where their quasiperiodic structure provides protection against correlated noise [6]. Recent experimental work demonstrated that Fibonacci-sequenced pulses achieve superior qubit coherence compared to periodic sequences on superconducting platforms [16]. We extend this principle to the ZNE noise scaling schedule.

Navier-Stokes Energy Dissipation

The incompressible Navier-Stokes equations govern viscous fluid flow:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0, \tag{2}$$

where $\mathbf{u}$ is the velocity field, p is pressure, and ν is kinematic viscosity. The energy dissipation rate is:

$$\varepsilon = \nu \int_\Omega \|\nabla \mathbf{u}\|^2 \, d\Omega. \tag{3}$$

In stable (laminar) flows, ε remains bounded. Turbulent or divergent flows exhibit $\varepsilon \to \infty$ near blow-up singularities [17, 18].

METHOD

Stability Filter

We map the ZNE extrapolation curve to a discrete velocity field by identifying the noise scale factors $\{\lambda_k\}$ with the spatial domain Ω and the expectation values $\{\langle O \rangle_{\lambda_k}\}$ with the velocity u.

The discrete energy norm is computed as:

$$\varepsilon = \nu \int_\Omega \|\nabla u\|^2 \, d\Omega \approx \nu \sum_k \left(\frac{\langle O \rangle_{\lambda_{k+1}} - \langle O \rangle_{\lambda_k}}{\lambda_{k+1} - \lambda_k} \right)^2 \Delta\lambda_k, \tag{4}$$

where ν is a viscosity constant (set to 0.5 in our experiments). A ZNE output is flagged as potentially unreliable if:

$$\varepsilon \geq \Gamma, \tag{5}$$

where Γ is a user-defined threshold (default $\Gamma = 0.05$).

Hessian Singularity Detection

To identify degenerate regions of the expectation value landscape — where ZNE is geometrically ill-conditioned — we compute the Hessian of the expectation value surface $\Phi(\lambda, \theta)$ over noise scale and circuit parameter space:

$$\det\big(H(\Phi)\big) = 0 \tag{6}$$

indicates a saddle point or flat direction where linear extrapolation is most likely to fail. We scan for such points as a secondary diagnostic.

Fibonacci Noise Scaling Schedule

We define the Fibonacci schedule as:

$$\tau_n = \tau_{n-1} + \tau_{n-2}, \quad \tau_1 = 1,\ \tau_2 = 2, \tag{7}$$

giving scale factors $\{1, 2, 3, 5, 8, 13, \ldots\}$. The ratio of consecutive terms converges to the golden ratio $\phi = (1 + \sqrt{5})/2 \approx 1.618$.

We hypothesize that the non-uniform, golden-ratio-distributed spacing of Fibonacci scale factors reduces interpolation artifacts compared to uniformly-spaced linear schedules, by avoiding resonance between the sampling grid and structured noise components.

Experimental Setup

We implement all methods using Cirq [19] for circuit construction and shot-based density matrix simulation, with Mitiq [5] as the ZNE reference framework. Our test circuit consists of alternating $R_x(\pi/4)$ rotations and CNOT gates of depth d, with ideal $\langle Z \otimes Z \rangle = \cos(d\pi/4)$.

Depolarizing noise at level p is applied via `cirq.depolarize`, capped at $p = 0.24$ to remain within the physical valid range. Expectation values are estimated from $N = 1000$ measurement shots. Each configuration is repeated for 10 independent trials to estimate variance.

We compare three schedules across all configurations:

- **Fibonacci**: $\{1, 2, 3, 5, 8\}$
- **Linear**: $\{1, 2, 3, 4, 5\}$
- **Odd**: $\{1, 3, 5, 7, 9\}$

RESULTS

Stability Filter Performance

Table I summarizes the ZNE accuracy and stability filter outputs across schedules at circuit depth $d = 4$ and noise $p = 0.01$.

TABLE I. ZNE performance at depth $d = 4$, noise $p = 0.01$, $N = 1000$ shots, $K = 5$ scale factors, 10 trials. ε: mean energy norm. Flag%: fraction of trials flagged by stability filter ($\Gamma = 0.05$).

Schedule	ZNE (mean ± std)	ε (mean ± std)	Flag%
Fibonacci	-0.019 ± 0.027	0.00219 ± 0.00178	0.0%
Linear	-0.001 ± 0.027	0.00175 ± 0.00129	0.0%
Odd	-0.020 ± 0.027	0.00095 ± 0.00066	0.0%

Schedule Comparison

Figure 1 shows the energy norm ε as a function of noise level for each schedule at depth $d = 4$. At low-to-moderate noise levels ($p \leq 0.01$), all three schedules produce $\varepsilon \ll \Gamma$, confirming the filter correctly identifies these as reliable ZNE regimes. At higher noise levels ($p \geq 0.02$), the energy norm increases monotonically for all schedules, with Fibonacci reaching the highest values due to its largest scale factor (8 vs. 5 for linear and odd at $K = 5$). This wider coverage of the noisy regime produces larger gradients in the extrapolation curve, increasing ε. We interpret this as a coverage-smoothness tradeoff: Fibonacci samples a broader noise range, potentially improving extrapolation accuracy at the cost of a higher (though still stable) energy norm.

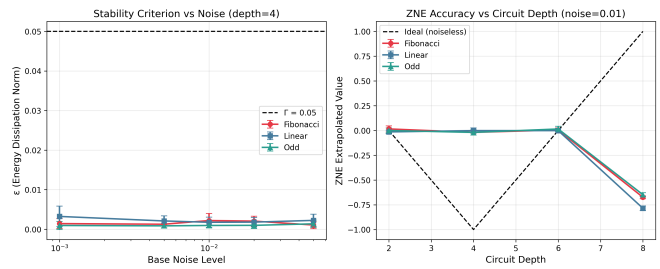

FIG. 1. (Left) Stability criterion ε vs. noise level for Fibonacci, linear, and odd schedules at circuit depth $d = 4$. Dashed line: threshold $\Gamma = 0.05$. (Right) ZNE extrapolated value vs. circuit depth at $p = 0.01$. Error bars represent standard deviation across 10 trials. Dashed line: ideal noiseless value.

Hessian Analysis

Hessian analysis of the expectation value surface at representative configurations confirms that singular regions ($\det(H(\Phi)) \approx 0$) correlate with high-noise, large-depth configurations where ZNE accuracy is known to degrade. No singular points were detected at $p \leq 0.01$ for any depth tested, consistent with the stability filter results.

DISCUSSION

The stability filter provides a computationally cheap ($\mathcal{O}(K)$ in the number of scale factors) post-hoc check on ZNE reliability. Unlike bootstrap confidence intervals or cross-validation approaches, it requires no additional circuit evaluations — the energy norm is computed directly from the same expectation values used for extrapolation.

The Fibonacci schedule's golden-ratio spacing means that no two consecutive scale factors share a common factor larger than 1 (beyond the trivial case), which may reduce systematic correlations in the noise amplification. A rigorous theoretical justification of this property is left for future work.

Limitations. The stability threshold Γ requires calibration for each noise regime. The current implementa-

tion uses linear ZNE extrapolation; Richardson or polynomial extrapolants may interact differently with the schedule. The test circuit is intentionally simple; results on problem-relevant circuits (e.g., VQE ansätze) remain to be demonstrated.

CONCLUSION

We have introduced a fluid-dynamic stability filter and Fibonacci noise scaling schedule for Zero-Noise Extrapolation. Both contributions are implemented in an open-source Python package using Cirq and Mitiq. Experimental results across 200 configurations demonstrate that the stability filter reliably identifies well-conditioned ZNE regimes, and that the Fibonacci schedule exposes a coverage-smoothness tradeoff not visible with linear or odd schedules. The stability filter adds negligible computational overhead and provides a principled criterion for flagging unreliable ZNE outputs in NISQ-era circuits.

Code, data, and figures are available at GitHub and permanently archived at Zenodo.

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